Plaza Works Job Management App

Module 11: KPI Tracking and Performance Management

Purpose

Track every metric that determines whether the business hits survival targets (Month 1) and growth targets (Months 2-6). Provide real-time visibility so problems are caught early and wins are recognized.

11.1 KPI Dashboard - Business Owner View

Daily Dashboard (Default View)

Today's Numbers - Per Tradesman

Metric	Target	Actual	Status
Labor revenue billed	$2,000	[live]	Green/Amber/Red
Hours logged	7.5 hrs	[live]	Green/Amber/Red
Jobs progressed/completed	1-2	[live]	Green/Amber/Red
Quotes sent (value)	$3,400	[live]	Green/Amber/Red

Status Thresholds:

- Green: At or above target
- Amber: 75-99% of target
- Red: Below 75% of target

Quick View Cards:

- Total labor revenue today (all tradesmen combined)
- Total quotes sent today (value)
- Jobs completed today
- Outstanding tasks requiring attention

Weekly Dashboard

Revenue Tracking - Per Tradesman

Metric	Target	Actual	Variance
Labor revenue billed	$10,000	[sum]	+/- $
Quotes sent (value)	$17,000	[sum]	+/- $
Quote acceptance rate	70%	[calculated]	+/- %
Days hitting $2k target	5	[count]	X/5

Team Overview:

- All tradesmen hitting labor KPI: [X]% of days
- Red flag if below 60%

Cash Position:

- Current bank balance
- Covers next X weeks wages (target: 2+ weeks)
- Warning if below 1 week coverage

Pipeline Health:

- Qualified leads in pipeline
- Sufficient for next X weeks (target: 2+ weeks)
- Warning if less than 1 week pipeline

Monthly Dashboard

Business Health Scorecard

Metric	Target	Actual	Status
Total revenue	[from plan]	[sum]	
Labor revenue (all tradesmen)	$43,300 x [team size]	[sum]	
Net profit	[from plan]	[calculated]	
Profit vs target	Within 10%	[calculated]	
Invoices outstanding >14 days	Under $10k	[sum]	
Jobs completed vs started	90%+	[calculated]	

Staff hitting KPIs	100%	[count]

Monthly Targets by Team Configuration:

Config	Revenue Target	Profit Target
1P + 1A	$64,950	$3,001
2P + 2A	$129,900	$22,328
3P + 2A	$194,850	$51,120

System auto-detects current team config and shows relevant targets.

Quarterly Dashboard

Metric	Target	Actual
Cumulative profit vs plan	On track	[sum]
Team growth vs plan	On track	[comparison]
Cash reserves trend	Increasing	[trend graph]

11.2 Role-Based Dashboards

Tradesman Dashboard

My Daily Numbers

Metric	Target	My Actual
Labor revenue billed	$2,000	[today's sum]
Hours logged	7.5 hrs	[today's sum]
Jobs progressed	1-2	[count]

My Weekly Numbers

Metric	Target	My Actual	Trend
Labor revenue	$10,000	[week sum]	↑↓
Quotes sent	$17,000	[week sum]	↑↓

Close rate	70%	[calculated]	↑↓

My Progress:

- Current sales phase: 1 / 2 / 3
- Weeks at current phase: [count]
- Progress to next phase: [checklist status]

Visual Elements:

- Progress bar showing weekly labor target
- Calendar heat map (green/amber/red per day)
- Streak counter for consecutive days hitting target

Apprentice Dashboard

My Daily Log

Metric	Target	My Actual
Hours logged	7.5 hrs	[today]
Assigned to tradesman	Yes	[status]

My Weekly Summary

Metric	Target	Actual
Total hours	37.5 hrs	[sum]
Unassigned time	0	[sum]
Tradesman hit target	Yes	[status]

Marketing Dashboard

Lead Generation - Weekly

Metric	Target	Actual
Drainage leads (qualified)	[scaled to team]	[count]
Bathroom leads (qualified)	[scaled to team]	[count]

Lead cost (drainage)	Under $50	[calculated]
Lead cost (bathroom)	Under $100	[calculated]
Lead response time	Under 2 hrs	[average]
Qualified lead handoff	Same day	[average]

Lead Targets by Team Size:

Team Size	Drainage Quotes/Week	Bathroom Leads/3 Weeks
1 Tradesman	$17,000	2
2 Tradesmen	$34,000	4
3 Tradesmen	$51,000	6

Monthly Trends:

- Cost per qualified lead trend
- Lead volume vs team capacity match
- Ad angles tested this month

11.3 Tradesman Sales Phase Tracking

Phase Definitions

Phase 1: Quoting Only

- Sends quotes
- Does not close sales independently
- Sales handled by Sales Coach or Business Owner

Phase 2: Quoting + Closing Drainage

- Closes drainage sales independently
- Bathroom sales still supported

Phase 3: Full Sales

- Closes all sales independently
- Drainage and bathroom

Phase Progression Tracker

Per Tradesman Record:

Name: [Tradesman Name]
Current Phase: [1 / 2 / 3]
Phase Start Date: [date]
Weeks at Phase: [calculated]
Current Close Rate: [calculated]

Phase 1 → 2 Checklist:

- [ ] 4+ weeks hitting $10k weekly labor
- [ ] Completed sales coaching sessions
- [ ] Demonstrated phone qualification skills
- [ ] Business Owner sign-off

Phase 2 → 3 Checklist:

- [ ] 8+ weeks at Phase 2
- [ ] $50k+ drainage jobs closed independently
- [ ] Shadowed 3+ bathroom sales
- [ ] Business Owner sign-off

Progression Dashboard:

- Visual timeline of each tradesman's progression
- Checklist completion status
- Ready/Not Ready indicator for next phase

Sales Bonus Calculator

Phase 2 Bonus Structure:

Weekly Closed Sales (Labor Value)	Bonus
$10,000 - $12,000	$0 (base KPI)
$12,001 - $14,000	$300
$14,001 - $16,000	$500
$16,001 - $18,000	$750
$18,001+	$1,000

Phase 3 Bonus Structure:

Weekly Closed Sales (Labor Value)	Bonus
$10,000 - $12,000	$0 (base KPI)
$12,001 - $15,000	$400
$15,001 - $18,000	$700
$18,001 - $22,000	$1,000
$22,001+	$1,500

Auto-Calculation:

- System tracks closed sales per tradesman per week
- Calculates bonus automatically
- Shows projected bonus in real-time during week
- Generates fortnightly bonus report for payroll

11.4 Profitability Tracking

Job-Level Profitability

Per Job Record:

Job: [Job ID]
Type: Drainage / Bathroom / Other
Tier: 1 / 2 / 3

Revenue:
- Labor billed: $X
- Materials billed: $X
- Total billed: $X

Costs:
- Materials cost: $X
- Labor cost: $X (calculated from hours x staff rate)
- Subcontractor cost: $X
- Overhead allocation: $X

Profit:
- Gross margin: $X
- Net profit: $X
- Daily profit rate: $X/day

Duration Tracking

Quoted vs Actual:

Metric	Quoted	Actual	Variance
Duration (days)	X	X	+/- X
Labor hours	X	X	+/- X

Efficiency Score:

- Jobs completed faster than quoted: [count]
- Jobs completed on time: [count]
- Jobs running over: [count]

Alert: Job running 1+ day over quoted duration

Backcosting Report

Purpose: Find discrepancy between staff-reported time and actual job revenue

Per Tradesman - Monthly:

Metric	Reported	Actual	Discrepancy
Hours logged	X	-	-
Jobs completed	X	X	-
Labor revenue	-	$X	-
Revenue per hour	-	$X	Compare to target
Expected hours (at target rate)	X	-	vs reported

Red Flag: Significant gap between reported hours and expected hours based on revenue

Daily Profit Visualization

Goal: Show $/hour/staff as a daily visual

Display:

- Each tradesman's daily revenue ÷ hours = $/hour
- Target: $266/hour ($2,000 ÷ 7.5 hours)
- Color coded: Green (above), Amber (close), Red (below)

Weekly Roll-Up:

- Average $/hour per tradesman
- Team average $/hour
- Trend over past 4 weeks

11.5 Pipeline and Forecasting

Quote Pipeline

Active Quotes:

Quote	Client	Value	Labor Value	Days Open	Status
Q-001	[name]	$X	$X	X	Sent/Followed Up

Pipeline Summary:

- Total quoted (open): $X
- Expected close (at 70%): $X
- Covers next X weeks of work

Booking Forecast

Schedule Capacity:

Period	Available Days	Booked Days	% Booked
This week	X	X	X%
Next week	X	X	X%
2 weeks out	X	X	X%
3 weeks out	X	X	X%
4 weeks out	X	X	X%

Target: Maintain 80%+ booking 2 weeks out

Alert: If booking drops below 60% for any period

Cash Flow Forecast

Weekly Projection:

Week	Expected Revenue	Expected Costs	Projected Balance
This week	$X	$X	$X
Next week	$X	$X	$X
Week 3	$X	$X	$X
Week 4	$X	$X	$X

Based On:

- Scheduled jobs (confirmed revenue)
- Pipeline at 70% close rate
- Known fixed costs
- Variable costs estimated from job mix

11.6 Alert System

Immediate Alerts (Push Notification + Dashboard)

Trigger	Alert To	Message
Daily labor under $1,500	Tradesman + Owner	"[Name] below daily target"
Free site visit logged	Owner	"Free site visit recorded - [Job]"
Cash below 1 week wages	Owner	"Cash reserve critical"
Job 1+ day over duration	Owner	"[Job] running over schedule"
Invoice 14+ days overdue	Owner	"[Invoice] overdue - $X"

Weekly Alerts (Dashboard + Email Summary)

Trigger	Alert To	Action
Tradesman <$8k labor for 2 weeks	Owner	Schedule review meeting

Close rate <50% for 2 weeks	Owner + Tradesman	Sales coaching focus
Lead cost >$80 (drainage)	Marketing	Review ad spend
Lead cost >$150 (bathroom)	Marketing	Review ad spend
Pipeline <1 week coverage	Owner + Marketing	Increase lead gen

Monthly Review Triggers

Condition	Flag
Missing monthly target by >10%	Course correction needed
Any tradesman stuck at Phase 1 for 3+ months	Progression review
Jobs running over duration regularly	Process review
Customer complaints increasing	Quality review

11.7 Reporting

Daily Report (Auto-Generated)

Sent to: Business Owner **Time:** End of day (6 PM)

Contents:

- Each tradesman: labor billed, hours logged, jobs completed
- Total team revenue today
- Any red flags triggered
- Tomorrow's scheduled work

Weekly Report (Auto-Generated)

Sent to: Business Owner, Sales Coach **Time:** Friday 5 PM

Contents:

- Per-tradesman KPI summary (all weekly metrics)
- Team totals
- Quote pipeline status
- Close rate analysis
- Cash position
- Schedule capacity next 4 weeks

- Red flags and action items

Monthly Report (Auto-Generated)

Sent to: Business Owner **Time:** 1st of month

Contents:

- Full P&L vs target
- Per-tradesman performance summary
- Phase progression updates
- Bonus calculations
- Trend analysis (improving/declining)
- Next month targets

Bonus Report (Auto-Generated)

Sent to: Business Owner **Time:** Fortnightly (payday)

Contents:

- Per-tradesman closed sales (labor value)
- Applicable bonus tier
- Bonus amount
- Pay period summary

11.8 Data Inputs Required

From Job Module

- Job created date
- Job type (drainage/bathroom/other)
- Quoted amount (total, labor, materials split)
- Quoted duration
- Actual completion date
- Actual hours logged
- Assigned tradesman
- Assigned apprentice

From Invoicing Module

- Invoice amounts
- Payment received dates
- Outstanding balances
- Days overdue

From Scheduling Module

- Booked days per tradesman
- Available capacity

From User Module

- Tradesman hourly cost (fully loaded)
- Apprentice hourly cost (fully loaded)
- Current sales phase per tradesman

From Quote Module

- Quote sent date
- Quote value
- Quote status (open/accepted/declined)
- Days to close

Manual Inputs

- Bank balance (or integrate with accounting)
- Ad spend (or integrate with ad platforms)
- Lead source and qualification status

11.9 KPI Configuration

Editable Targets

Business Owner can adjust:

Setting	Default	Editable
Daily labor target	$2,000	Yes
Weekly labor target	$10,000	Yes
Close rate target	70%	Yes
Lead cost threshold (drainage)	$50	Yes
Lead cost threshold (bathroom)	$100	Yes
Schedule booking target	80%	Yes
Cash reserve target (weeks)	2	Yes

Team Configuration

Set current team structure:

- Number of plumbers: [1/2/3+]
- Number of apprentices: [1/2+]
- System auto-calculates targets based on config

Bonus Structure

Editable bonus tiers:

- Phase 2 thresholds and amounts
- Phase 3 thresholds and amounts

11.10 Historical Analysis

Trend Views

Available Timeframes:

- Last 7 days
- Last 4 weeks
- Last 3 months
- Last 12 months
- Custom date range

Trend Metrics:

- Labor revenue (per tradesman and total)
- Close rate
- Average job profit
- Average job duration variance
- Lead cost
- Cash position

Comparison Views

Compare:

- Tradesman vs tradesman
- This month vs last month
- This quarter vs last quarter
- Actual vs target

Export Options

- CSV export for any report
- PDF export for monthly summaries
- API access for external tools (Power BI, etc.)

Integration with Existing Modules

This section details exactly how the KPI module connects to each module from the main feature specification.

Integration with Module 1: User Management

Data the KPI Module Reads:

Field	Source Table	Used For
user_id	users	Linking all KPIs to specific staff
role	users	Determining which dashboard to show
employment_type	users	Filtering permanent vs contractor stats
hourly_cost	users (custom field)	Calculating job profitability
daily_cost	users (custom field)	Calculating daily profit targets
sales_phase	users (custom field)	Showing phase progression, bonus tier
phase_start_date	users (custom field)	Calculating weeks at current phase

Data the KPI Module Writes:

Field	Target Table	Triggered By
sales_phase	users	Business Owner approval of phase progression
phase_start_date	users	Phase change

phase_history	user_phase_log (new)	Any phase change (audit trail)

New Fields Required in Users:

hourly_cost_loaded: decimal (fully loaded hourly rate)
daily_cost_loaded: decimal (fully loaded daily rate)
sales_phase: integer (1, 2, or 3)
phase_start_date: date
weekly_labor_target: decimal (default $10,000)
daily_labor_target: decimal (default $2,000)

New Table: user_phase_log

id: primary key
user_id: foreign key to users
previous_phase: integer
new_phase: integer
changed_by: foreign key to users
changed_at: timestamp
notes: text

Events Listened To:

- user.created → Initialize KPI tracking for new staff
- user.role_changed → Update dashboard permissions
- user.deactivated → Archive KPI history, exclude from active reports

Events Published:

- user.phase_changed → Triggers notification to user and owner

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Integration with Module 3: Job Management

Data the KPI Module Reads:

Field	Source Table	Used For
job_id	jobs	Linking revenue to specific jobs
client_id	jobs	Client-level reporting
job_type	jobs	Drainage vs bathroom breakdown

status	jobs	Counting completed vs in-progress
created_at	jobs	Timeline analysis
completed_at	jobs	Duration calculations
assigned_user_id	jobs	Per-tradesman revenue
assigned_apprentice_id	jobs	Apprentice utilization
quoted_duration_days	jobs	Duration variance tracking
actual_duration_days	jobs (calculated)	Duration variance tracking
quoted_labor_amount	jobs	Expected revenue
quoted_materials_amount	jobs	Materials pass-through tracking
tier	jobs	Tier distribution analysis

Calculations Performed:

actual_duration_days = completed_at - start_date (excluding non-work days)
duration_variance = actual_duration_days - quoted_duration_days
job_labor_revenue = sum of labor line items from invoices
job_profit = job_labor_revenue - (hours_logged x assigned_user.hourly_cost_loaded)
daily_profit_rate = job_profit / actual_duration_days

Events Listened To:

- job.created → Add to pipeline tracking
- job.status_changed → Update completion counts, trigger duration check if completed
- job.completed → Calculate final profitability, update tradesman revenue totals
- job.assigned → Link job to tradesman KPIs

Aggregations Built:

Aggregation	Grouping	Frequency
daily_labor_revenue	user_id, date	Real-time
weekly_labor_revenue	user_id, week	Real-time
monthly_labor_revenue	user_id, month	Real-time
jobs_completed_count	user_id, period	Real-time

average_job_duration_variance	user_id, month	Daily recalc

Integration with Module 4: Scheduling

Data the KPI Module Reads:

Field	Source Table	Used For
schedule_id	schedules	Linking to capacity
user_id	schedules	Per-tradesman capacity
job_id	schedules	Confirmed vs tentative work
scheduled_date	schedules	Capacity by period
duration_hours	schedules	Hours booked
status	schedules	Confirmed vs tentative

Calculations Performed:

```
available_days_per_week = 5 (configurable)
booked_days = count of scheduled_date where status = confirmed
percent_booked = booked_days / available_days_per_week x 100

For each period (this week, next week, 2 weeks, 3 weeks, 4 weeks):
  - Count booked days per tradesman
  - Calculate team total capacity
  - Flag if below 60% threshold
```

Capacity Dashboard Query:

```
For user X, for period Y:
  SELECT COUNT(DISTINCT scheduled_date) as booked_days
  FROM schedules
  WHERE user_id = X
  AND scheduled_date BETWEEN [period_start] AND [period_end]
  AND status = 'confirmed'
```

Events Listened To:

- schedule.created → Update capacity forecast

- schedule.updated → Recalculate capacity
- schedule.cancelled → Recalculate capacity, flag if drops below threshold

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Integration with Module 6: Quotes, Invoices and Payments

Quote Data Read:

Field	Source Table	Used For
quote_id	quotes	Pipeline tracking
job_id	quotes	Linking to jobs
client_id	quotes	Client analysis
total_amount	quotes	Pipeline value
labor_amount	quotes	Labor-specific tracking
materials_amount	quotes	Materials pass-through
status	quotes	Open/accepted/declined counts
sent_at	quotes	Days open calculation
responded_at	quotes	Close time tracking
created_by	quotes	Per-tradesman quote volume

Invoice Data Read:

Field	Source Table	Used For
invoice_id	invoices	Revenue tracking
job_id	invoices	Linking to jobs
total_amount	invoices	Revenue totals
labor_amount	invoices	Labor revenue (KPI basis)
status	invoices	Paid vs outstanding
due_date	invoices	Aging analysis

paid_at	invoices	Cash flow timing
days_overdue	invoices (calculated)	Overdue alerts

Payment Data Read:

Field	Source Table	Used For
payment_id	payments	Cash received tracking
invoice_id	payments	Linking to invoices
amount	payments	Cash in
received_at	payments	Cash flow timing

Calculations Performed:

Quote Pipeline:
open_quotes_value = SUM(total_amount) WHERE status = 'sent'
open_quotes_labor = SUM(labor_amount) WHERE status = 'sent'
expected_close_value = open_quotes_value x 0.70

Close Rate (per tradesman, per period):
quotes_sent = COUNT WHERE created_by = user_id AND sent_at IN period
quotes_accepted = COUNT WHERE created_by = user_id AND status = 'accepted' AND responded_at IN period
close_rate = quotes_accepted / quotes_sent x 100

Days to Close:
average_days_to_close = AVG(responded_at - sent_at) WHERE status = 'accepted'

Outstanding Invoices:
overdue_amount = SUM(amount_due) WHERE status != 'paid' AND due_date < today
overdue_14_plus = SUM(amount_due) WHERE days_overdue > 14

Events Listened To:

- quote.sent → Add to pipeline, start tracking
- quote.accepted → Update close rate, remove from pipeline
- quote.declined → Update close rate, remove from pipeline
- invoice.created → Add to revenue tracking
- invoice.paid → Update cash position, remove from outstanding
- invoice.overdue → Trigger alert if over threshold
- payment.received → Update cash position

Integration with Module 7: Workflow and Checklists

Data the KPI Module Reads:

Field	Source Table	Used For
task_id	tasks	Task completion tracking
job_id	tasks	Linking to jobs
assigned_to	tasks	Per-user task counts
completed_at	tasks	Completion tracking
hours_logged	tasks	Time tracking

Time Logging Integration:

The KPI module relies heavily on accurate time logging from the workflow module.

```
Daily Hours Query:
  SELECT SUM(hours_logged) as total_hours
  FROM tasks
  WHERE assigned_to = user_id
  AND DATE(completed_at) = target_date

Weekly Hours Query:
  SELECT SUM(hours_logged) as total_hours
  FROM tasks
  WHERE assigned_to = user_id
  AND completed_at BETWEEN week_start AND week_end
```

Events Listened To:

- task.completed → Update hours logged, recalculate daily/weekly totals
- task.hours_updated → Recalculate totals

End-of-Day Check:

```
At 6 PM daily:
  For each active tradesman:
    IF hours_logged_today < 6 hours:
      Trigger reminder notification
    IF hours_logged_today = 0:
      Trigger escalation to Business Owner
```

Integration with Module 8: Notifications and Communication

Alerts the KPI Module Triggers:

Alert Type	Condition	Channel	Recipients
daily_target_warning	Labor < $1,500	Push + In-app	Tradesman
daily_target_missed	Labor < $1,500 at EOD	Email	Owner
free_site_visit	Site visit with $0 charge	Push	Owner
cash_critical	Balance < 1 week wages	SMS + Email	Owner
job_over_duration	Actual > Quoted + 1 day	In-app	Owner
invoice_overdue	Days overdue > 14	Email	Owner
weekly_target_missed	Labor < $8,000 for week	Email	Owner + Tradesman
close_rate_low	Close rate < 50% for 2 weeks	Email	Owner + Tradesman
pipeline_low	Pipeline < 1 week coverage	Push + Email	Owner + Marketing
lead_cost_high	Drainage > $80 or Bath > $150	Email	Marketing
phase_ready	All checklist items complete	In-app	Owner
bonus_earned	Closed sales exceed threshold	In-app	Tradesman

Notification Payload Format:

```
{
  "alert_type": "daily_target_warning",
  "severity": "warning",
  "title": "Below daily target",
  "message": "Current labor revenue: $1,234. Target: $2,000.",
  "data": {
    "user_id": "user_123",
    "current_value": 1234,
    "target_value": 2000,
    "gap": 766
  },
```

```
  "channels": ["push", "in_app"],
  "recipients": ["user_123"]
}
```

Events Published:

- kpi.alert_triggered → Notification module sends appropriate message
- kpi.daily_report_ready → Notification module sends daily summary
- kpi.weekly_report_ready → Notification module sends weekly summary

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Integration with Module 9: Reporting and Analytics

The KPI module extends the existing reporting module with specific KPI reports.

New Report Types:

Report	Frequency	Format	Distribution
daily_kpi_summary	Daily 6 PM	In-app + Email	Owner
weekly_kpi_report	Friday 5 PM	PDF + Email	Owner, Sales Coach
monthly_business_review	1st of month	PDF + Email	Owner
tradesman_performance	Weekly	In-app	Each tradesman (own data)
bonus_calculation	Fortnightly	PDF	Owner (for payroll)
pipeline_status	Daily	In-app	Owner, Marketing
backcosting_analysis	Monthly	PDF + Email	Owner

Report Data Sources:

```
Daily KPI Summary:
  - kpi_daily_snapshots table (see new tables below)
  - jobs (today's completions)
  - tasks (today's hours)
  - quotes (today's sent/accepted)

Weekly KPI Report:
  - kpi_weekly_snapshots table
  - Aggregated from daily snapshots
  - Pipeline current state
  - Schedule capacity
```

```
Monthly Business Review:
  - kpi_monthly_snapshots table
  - P&L from invoicing module
  - Comparison to targets from kpi_targets table
```

Integration with Module 10: Daily Operations

Staff Daily Tasks Integration:

The KPI module adds mandatory daily tasks based on role:

Tradesman Daily Tasks (Auto-Added):

- Log hours for today
- Update job progress
- Submit end-of-day photos (if job worked)

Apprentice Daily Tasks (Auto-Added):

- Log hours for today
- Confirm tradesman assignment

Business Owner Daily Tasks (Auto-Added):

- Review daily KPI dashboard
- Address any red flags
- Approve/action any pending items

Completion Tracking:

```
IF tradesman has not completed "Log hours" task by 6 PM:
  Send reminder notification

IF tradesman has not completed "Log hours" task by 8 PM:
  Flag in next day's Owner dashboard

IF pattern continues 3+ days:
  Escalate as "Staff not logging hours" alert
```

New Database Tables for KPI Module

kpi_daily_snapshots

```
id: primary key
user_id: foreign key to users
```

snapshot_date: date
labor_revenue: decimal
hours_logged: decimal
jobs_completed: integer
quotes_sent_value: decimal
quotes_sent_count: integer
quotes_accepted_value: decimal
quotes_accepted_count: integer
target_labor: decimal
target_met: boolean
created_at: timestamp

kpi_weekly_snapshots

id: primary key
user_id: foreign key to users
week_start: date
week_end: date
labor_revenue: decimal
hours_logged: decimal
jobs_completed: integer
quotes_sent_value: decimal
close_rate: decimal
target_labor: decimal
target_met: boolean
days_target_met: integer
created_at: timestamp

kpi_monthly_snapshots

id: primary key
user_id: foreign key to users (nullable for team totals)
month: date (first of month)
labor_revenue: decimal
materials_revenue: decimal
total_revenue: decimal
total_costs: decimal
net_profit: decimal
target_revenue: decimal
target_profit: decimal
variance_revenue: decimal
variance_profit: decimal
jobs_completed: integer
jobs_started: integer
average_close_rate: decimal
created_at: timestamp

kpi_targets

id: primary key
team_config: string (e.g., "1P+1A", "2P+2A", "3P+2A")
daily_labor_target: decimal
weekly_labor_target: decimal
monthly_revenue_target: decimal
monthly_profit_target: decimal
close_rate_target: decimal
is_active: boolean
effective_from: date
created_at: timestamp
updated_at: timestamp

kpi_alerts_log

id: primary key
alert_type: string
severity: string (info, warning, critical)
user_id: foreign key (who it's about)
triggered_at: timestamp
message: text
data: json
acknowledged: boolean
acknowledged_by: foreign key
acknowledged_at: timestamp

tradesman_bonus_periods

id: primary key
user_id: foreign key
period_start: date
period_end: date
closed_sales_labor_value: decimal
sales_phase: integer
bonus_tier: string
bonus_amount: decimal
approved_by: foreign key
approved_at: timestamp
paid: boolean
paid_at: timestamp

phase_progression_checklist

id: primary key
user_id: foreign key
from_phase: integer
to_phase: integer
checklist_item: string
completed: boolean
completed_at: timestamp
verified_by: foreign key
notes: text

API Endpoints for KPI Module

Dashboard Endpoints:

GET /api/v1/kpi/dashboard/daily
- Returns: Today's KPIs for requesting user's role
- Params: date (optional, defaults to today)

GET /api/v1/kpi/dashboard/weekly
- Returns: This week's KPIs
- Params: week_start (optional)

GET /api/v1/kpi/dashboard/monthly
- Returns: This month's KPIs
- Params: month (optional)

GET /api/v1/kpi/dashboard/owner
- Returns: Full owner dashboard with all tradesmen
- Requires: Owner or Admin role

Tradesman Endpoints:

GET /api/v1/kpi/tradesman/{user_id}/summary
- Returns: KPI summary for specific tradesman
- Requires: Owner role or own user_id

GET /api/v1/kpi/tradesman/{user_id}/phase
- Returns: Current phase, progression status, checklist

POST /api/v1/kpi/tradesman/{user_id}/phase/advance
- Action: Move tradesman to next phase
- Requires: Owner role
- Validates: All checklist items complete

Pipeline Endpoints:

GET /api/v1/kpi/pipeline
- Returns: Quote pipeline summary

GET /api/v1/kpi/capacity
- Returns: Schedule capacity for next 4 weeks

Reporting Endpoints:

GET /api/v1/kpi/reports/daily
- Returns: Daily report data
- Params: date

GET /api/v1/kpi/reports/weekly
- Returns: Weekly report data
- Params: week_start

GET /api/v1/kpi/reports/monthly
- Returns: Monthly report data
- Params: month

POST /api/v1/kpi/reports/export
- Action: Generate PDF/CSV export
- Params: report_type, format, period

Alert Endpoints:

GET /api/v1/kpi/alerts
- Returns: Active alerts for user
- Params: acknowledged (filter)

POST /api/v1/kpi/alerts/{alert_id}/acknowledge
- Action: Mark alert as acknowledged

Bonus Endpoints:

GET /api/v1/kpi/bonus/current
- Returns: Current period bonus projection for user

GET /api/v1/kpi/bonus/history
- Returns: Past bonus periods

POST /api/v1/kpi/bonus/{period_id}/approve
- Action: Approve bonus for payment
- Requires: Owner role

Event Flow Diagrams

Daily Revenue Tracking Flow:

```
1. Tradesman completes task with hours logged
   ↓
2. task.completed event fires
   ↓
3. KPI module listens, updates kpi_daily_snapshots
   ↓
4. If labor_revenue crosses threshold:
   - Below $1,500 → trigger daily_target_warning alert
   - Above $2,000 → update target_met = true
   ↓
5. At 6 PM, daily snapshot finalized
   ↓
6. kpi.daily_report_ready event fires
   ↓
7. Notification module sends daily summary
```

Quote Close Rate Flow:

```
1. Quote status changes to 'accepted' or 'declined'
   ↓
2. quote.accepted or quote.declined event fires
   ↓
3. KPI module listens, recalculates close rate for user
   ↓
4. Updates kpi_weekly_snapshots.close_rate
   ↓
5. If close_rate < 50% for 2 consecutive weeks:
   ↓
6. Trigger close_rate_low alert
```

Phase Progression Flow:

```
1. Owner views tradesman's phase progression screen
   ↓
2. System shows checklist from phase_progression_checklist
   ↓
3. Each item checked/unchecked updates the record
   ↓
4. When all items complete:
   ↓
```

5. "Advance Phase" button becomes active
   ↓
6. Owner clicks advance, confirms
   ↓
7. POST /api/v1/kpi/tradesman/{id}/phase/advance
   ↓
8. System updates users.sales_phase, creates audit log
   ↓
9. user.phase_changed event fires
   ↓
10. Notification sent to tradesman

Scheduled Jobs

Job	Frequency	Time	Action
daily_snapshot_finalize	Daily	6:00 PM	Finalize daily KPI snapshots
daily_report_generate	Daily	6:05 PM	Generate and send daily reports
weekly_snapshot_generate	Weekly	Friday 5:00 PM	Generate weekly snapshots
weekly_report_generate	Weekly	Friday 5:05 PM	Generate and send weekly reports
monthly_snapshot_generate	Monthly	1st, 12:00 AM	Generate monthly snapshots
monthly_report_generate	Monthly	1st, 6:00 AM	Generate and send monthly reports
hours_reminder	Daily	6:00 PM	Send reminder to staff who haven't logged hours
alert_escalation_check	Hourly	Every hour	Check for alerts needing escalation
bonus_period_close	Fortnightly	Payday -2 days	Calculate and flag bonus for approval

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Build Priority for KPI Module

Phase 1: Essential Tracking

- Daily/weekly labor revenue per tradesman
- Hours logging
- Basic dashboard (daily numbers)

Phase 2: Financial Visibility

- Quote pipeline tracking
- Close rate calculation
- Monthly P&L summary
- Cash position tracking

Phase 3: Alerts and Automation

- Red flag alert system
- Auto-generated reports
- Schedule capacity view

Phase 4: Advanced Analytics

- Backcosting comparison
- $/hour visualization
- Trend analysis
- Tradesman comparison

Phase 5: Sales Phase Management

- Phase progression tracking
- Bonus calculation
- Phase checklists

Success Criteria

The KPI module is working when:

1. Business Owner can see at a glance if today/this week is on target
2. Every tradesman knows their numbers without asking
3. Problems surface within 24 hours, not at month end
4. Bonus calculations are automatic and accurate
5. No one can claim they "didn't know" they were behind target
6. Cash surprises are eliminated through forecasting